DETERMINATION OF Z PARAMETERS OF TWO PORT NETWORK

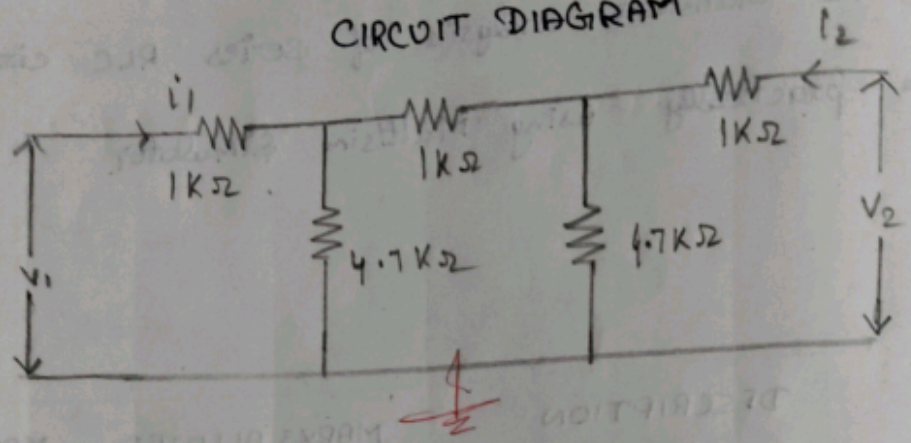
AIM :-

To calculate and verify Z parameters of a two port network using multisim simulator

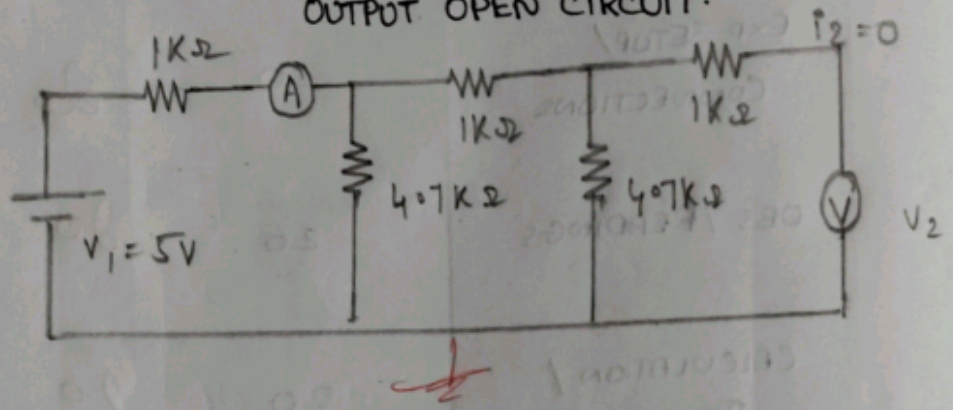
Apparatus required :-

S.no	Components	specification	Quantity
1	power supply		1
2	multimeter	---	1
3	Resistors	10kΩ, 10kΩ, 4.7kΩ	
4	bread board	---	1
5	Connecting wires	---	---
6	PC with Multisim simulator	---	---

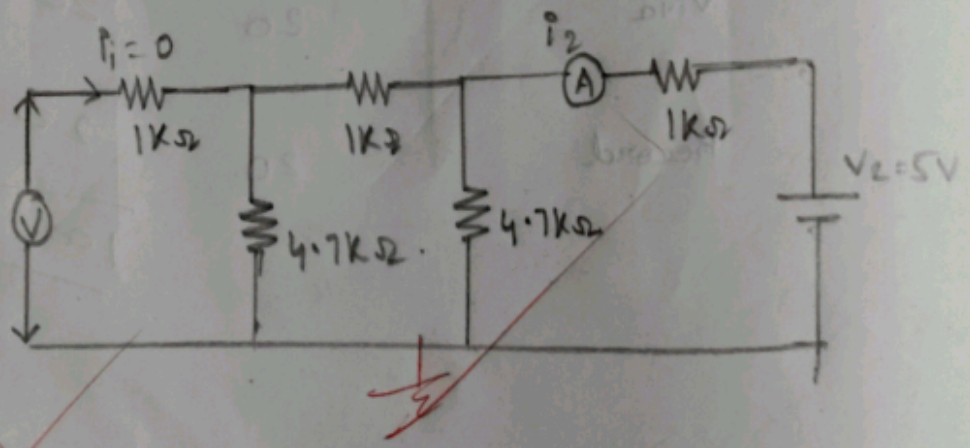
CIRCUIT DIAGRAM



OUTPUT OPEN CIRCUIT.



INPUT OPEN CIRCUITED :-



PROCEDURE :-

(a) Connect the circuit as shown in fig. & switch 'ON' the experimental board.

(b) First open the O/P terminal & supply 5V to I/P terminal. Measure O/P voltage & I/P Current.

(c) Secondly, open I/P terminal & supply 5V to O/P terminal

(d) Calculate the values of z parameter using Eq ① and ②

(e) Switch 'OFF' the supply after taking the readings.

OBSERVATION

output open circuit			$Z_{21} = V_2/I_1$ (Ω)	$Z_{11} = V_1/I_1$ (Ω)
V_1	V_2	I_1		
5.0	2.60	2.10	1.23 KΩ	2.38 KΩ
5.0	2.55	2.08	1.22 KΩ	2.40 KΩ
5.0	2.63	2.12	1.24 KΩ	2.35 KΩ

Input open circuit			$Z_{12} = V_1/I_2$ (Ω)	$Z_{22} = V_2/I_2$ (Ω)
V_2	V_1	I_2		
5.0	2.60	2.10	1.23	4.00
5.0	2.57	2.09	1.22	2.39
5.0	2.62	2.12	1.24	2.38

Verified

Result :-

The z-parameter of the two port network has been calculated and verified by using multisim simulator.

SL.NO	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1	EXP SETUP/ CONNECTIONS	20	20
2	OBS / READINGS	20	20
3	CALCULATION	20	20
4	Viva	20	20
5	Record	20	20
			100